Three papers from one common theory: complete proofs and measured costs

Human Compression project

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The specified main results of three historical manuscripts follow from one common theory and three complete applications. Counting the full source manuscripts and the full replacement gives

$$R_E = \frac{c(L083) + c(L079) + c(L021)}{c(U_E) + c(A_1) + c(A_2) + c(A_3)} = \frac{71{,}375}{39{,}303} = \boxed{1.816019\ldots}.$$

Here c counts NFC characters other than whitespace in the bound PDF extractions. The entire common theory is paid once in this topic. Any overlap with another topic does not reduce that charge.

Once the common theory and its required program are available, the complete applications have the following conditional factors:

Source / application	Full source	Complete application	Conditional factor
L083 / PSL polytopes	20,885	3,002	6.957029
L079 / PGL polytopes	20,908	1,855	11.271159
L021 / PSL maniplexes	29,582	14,656	2.018422
Shared theory	—	19,790	paid once
Full topic	71,375	39,303	1.816019

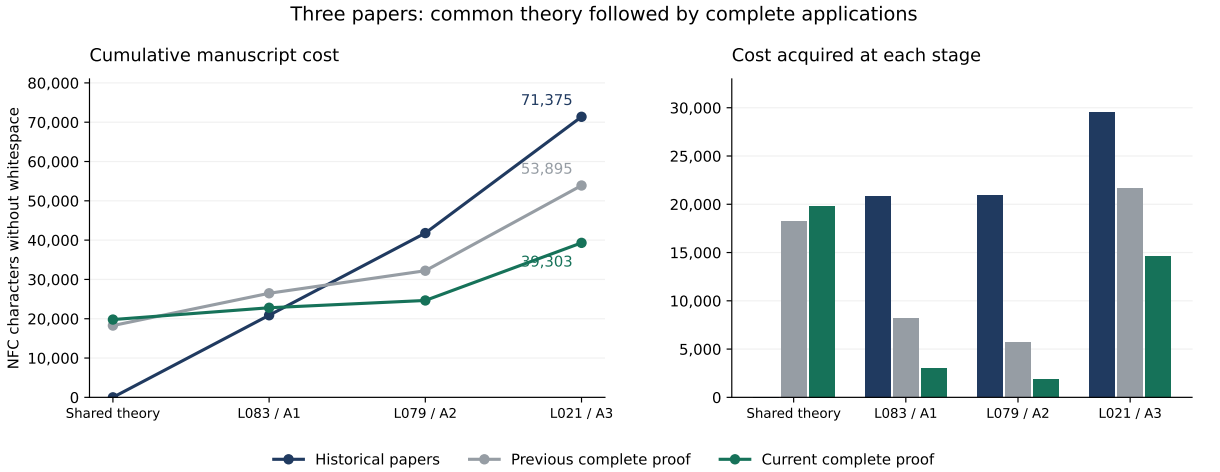
Each conditional factor is $c(L_i)/c(A_i)$ given the complete shared theory and program. It does not describe the cost of proving that application independently. No introductory text, proof, reference or required local argument has been removed from the application cost.

The sources are Leemans–Schulte's *Groups of type $L_2(q)$ acting on polytopes* and *Polytopes with groups of type $PGL_2(q)$*, and Leemans–Toledo's *Maniplexes with automorphism group $PSL(2,q)$*. The exact author editions and results recovered are fixed in the protocol. L079's semilinear census and conjecture, and L021's final PGL question, are excluded from those results; their printed source text is nevertheless counted. The previously reviewed corrections to L021 remain included.

The common theorem concerns four distinct string involutions whose two end groups are proper and isomorphic to S_4 or A_5, with both outer labels odd. It derives the three exact marked groups, their intersection property and uniqueness. One proof of the projective reduction, one finite calculation and one geometric argument serve all three applications. The maniplex application also uses the common extension theorem and proves its complete families for the stated ranges of primes. The historical local classes and universal presentations remain attributed to the earlier work. No additional source, conjecture solution or historical priority is claimed.

Required programs and the complete cost record. The complete required program has 7,565 characters. Thus the denominator including it is 46,868, and the corresponding source-text factor is 1.522894…. Including the complete diagnostic program adds 5,173 characters, giving denominator 52,041 and factor 1.371515…. Historical program costs and the full costs of the classical results are unknown. These additional measurements therefore do not claim to be exact ratios for every artifact on both sides. The same declared classical background is available to the historical and replacement proofs.

The new required and diagnostic programs together have 12,738 characters, 139 more than the previous programs. The improvement in the written proofs persists when the diagnostics are kept. Available Lean certificates verify only their stated supporting lemmas; the cost of a complete formalization remains unknown.



Written manuscripts in full; umbrella paid once. Required programs and diagnostic sensitivity are tabulated separately.

The earlier complete proof used 53,895 written characters. The current proof uses 39,303, a reduction of 14,592. The first edition of the common proof, preserved at commit `6d203fd`, used 38,568 and had factor 1.850627…. The readable edition adds 735 characters for motivation and clearer prose; it preserves the same mathematics and all required programs. Its figures above supersede the earlier figures for the current text, without changing the historical record.

The plotted order pays for the whole common theory before the complete applications. It is an accounting order, not a reconstruction of discovery or evidence of a repeated sawtooth. The CSV tables include every declared metric, the conditional factors, and all program costs.

The two independent implementations agree on all 150 document–metric values. Private and public replays produce identical result bytes. Ten negative controls reject changes to inputs, boundaries, glyphs, program scope and the recorded prose edits. Only individually bound page footers are removed from the new manuscripts; no proof text is omitted.